ELTE-IK Computer Science BSc, Basic Mathematics, 2020 autumn
6. Order preserving estimates
Solutions of exercises 1a., 1b., 1c., 2a., 2b., 2c., 3a of the section "Exercises for class work"
7. Statements, logical operations
Solutions of exercises 3, 5, 6, 7b., 7c of the section "Exercises for class work"
(author: Bóka Dávid)

6.2.1. Exercises for class work / 1a.

We know, that if $x > 0$, then $3x^4, 2x^2, 5 > 0$, that is

$$P(x) = 4x^5 - 3x^4 - 2x^2 - 5 = 4x^5 - \underbrace{(3x^4 + 2x^2 + 5)}_{>0} < 4x^5.$$

Based on this, $M = 4$ and, for example, $R = 1$ (any positive real number can be chosen here) are good parameters for the OPU-estimation.

6.2.1. Exercises for class work / 1b.

We know, that if $x > 0$, then $3x^2 > 0$, that is

$$P(x) = 2x^3 - 3x^2 + 6x + 7 < 2x^3 + 6x + 7.$$

On the other hand, if $x \geq 1$ and $n \geq 0$, then $\forall i \in \{0, 1, \ldots, n-1\}$: $x^i \leq x^n$, so

$$2x^3 + 6x + 7 \leq 2x^3 + 6x^3 + 7x^3 = 15x^3.$$

Based on this, $M = 15$ and $R = 1$ are good parameters for the OPU-estimation.

6.2.1. Exercises for class work / 1c.

Similar to the previous section if $x \geq 1$ and $n \geq 0$, then $\forall i \in \{0, 1, \ldots, n-1\}$: $x^i \leq x^n$, that is

$$P(x) = 6x^5 + 7x^4 + 10x^3 + x^2 + 2x + 3 \leq 6x^5 + 7x^5 + 10x^5 + x^5 + 2x^5 + 3^5 = 27x^5.$$

Based on this, $M = 27$ and $R = 1$ are good parameters for the OPU-estimation.

6.2.1. Exercises for class work / 2a.

We know, that if $x > 0$, then $7x^4, 10x^3, x^2, 2x, 3 > 0$, that is

$$P(x) = 6x^5 + 7x^4 + 10x^3 + x^2 + 2x + 3 > 6x^5.$$

Based on this, $M = 6$ and, for example, $R = 1$ (any positive real number can be chosen here) are good parameters for the OPL-estimation.

$\boxed{\text{6.2.1. Exercises for class work / 2b.}}$

We know, that if $x > 0$, then $6x, 7 > 0$, that is

$$P(x) = 2x^3 - 3x^2 + 6x + 7 > 2x^3 - 3x^2.$$

We can no longer estimate the difference on the right side in such a simple way, so we use the following idea:

$$2x^3 - 3x^2 = x^3 + x^3 - 3x^2 = x^3 + x^2(x-3) \geq x^3,$$

because, if $x \geq 3$, then the term $x^2(x-3)$ is non-negative. Based on this, $M = 1$ and $R = 3$ are good parameters for the OPL-estimation.

Remark: We can use another decomposition of the term $2x^3$ to get a good OPL-estimation, but with other parameters. For example:

$$2x^3 - 3x^2 = \frac{3}{2}x^3 + \frac{1}{2}x^3 - 3x^2 = \frac{3}{2}x^3 + x^2\left(\frac{1}{2}x - 3\right) \geq \frac{3}{2}x^3.$$

since, if $x \geq 6$, then the term $x^2\left(\frac{1}{2}x - 3\right)$ is non-negative. In this case $m = \frac{3}{2}$ and $R = 6$ are good parameters for the OPL-estimation.

$\boxed{\text{6.2.1. Exercises for class work / 2c.}}$

$$P(x) = 4x^5 - 3x^4 - 2x^2 - 5 = 4x^5 - (3x^4 + 2x^2 + 5)$$

One difference can be lower-estimated by upper-estimating the subtrahend, i.e. we produce the OPU-estimate of $3x^4 + 2x^2 + 5$:

$$3x^4 + 2x^2 + 5 \leq 3x^4 + 2x^4 + 5x^4 = 10x^4,$$

if $x \geq 1$. Using this, we can further estimate the polynomial $P(x)$ as follows:

$$4x^5 - (3x^4 + 2x^2 + 5) \geq 4x^5 - 10x^4 = 3x^5 + x^5 - 10x^4 = 3x^5 + x^4(x - 10) \geq 3x^5,$$

since, if $x \geq 10$, then the term $x^4(x-10)$ is non-negative. Based on this $m = 3$ and $R = \max(1, 10) = 10$ are good parameters for the OPL-estimation.

Remark: We can use another decomposition of the term $4x^5$. For example:

$$4x^5 - (3x^4 + 2x^2 + 5) \geq 4x^5 - 10x^4 = 2x^5 + 2x^5 - 10x^4 = 2x^5 + x^4(2x - 10) \geq 2x^5.$$

Here the term $x^4(2x-10)$ is non-negative, if $x \geq 5$, so in this case $m = 2$ and $R = 5$ are good parameters for the OPL-estimation.

$\boxed{\text{6.2.1. Exercises for class work / 3a.}}$

In this problem we have to give a lower and an upper estimate for a rational function. It is important to note that if we want to estimate a fraction

- from below, we must lower-estimate the numerator and / or upper-estimate the denominator.
- from above, we must upper-estimate the numerator and / or lower-estimate the denominator.

Based on the above, the OPL estimate of $f(x)$ requires the OPL estimate of the polynomial in the numerator:

$$3x^4 + 2x^3 + 5x^2 + 7x + 6 > 3x^4, \text{ for all positive real numbers x.}$$

And the OPU-estimate of the polynomial in the denominator:

$$5x^2 - 3x - 10 < 5x^2, \text{ for all positive real numbers x.}$$

Using these we get:

$$f(x) = \frac{3x^4 + 2x^3 + 5x^2 + 7x + 6}{5x^2 - 3x - 10} < \frac{3x^4}{5x^2} = \frac{3}{5}x^2.$$

Based on this, $m = \frac{3}{5}$ and, for example, $R = 1$ (any positive real number can be chosen here) are good parameters for the OPL-estimation.

For the OPU-estimation of $f(x)$ we need the OPU-estimate of the polynomial in the numerator:

$$3x^4 + 2x^3 + 5x^2 + 7x + 6 \le 3x^4 + 2x^4 + 5x^4 + 7x^4 + 6x^4 = 23x^4, \text{ if } x \ge 1.$$

And the OPL-estimate of the polynomial in the denominator:

$$5x^2 - 3x - 10 = 5x^2 - (3x + 10) > 5x^2 - 13x = 4x^2 + x^2 - 13x = 4x^2 + x(x - 13) \ge 4x^2, \text{ if } x \ge 13.$$

Using these we get:

$$f(x) = \frac{3x^4 + 2x^3 + 5x^2 + 7x + 6}{5x^2 - 3x - 10} \ge \frac{23x^4}{4x^2} = \frac{23}{4}x^2.$$

Based on this, $M = \frac{23}{4}$ and $R = \max(1, 13) = 13$ are good parameters for the OPU-estimation.

7.2.1. Exercises for class work / 3a.

$x = 0$ és $y = 0 \Longrightarrow x^2 + y^2 = 0$. True, because $0^2 + 0^2 = 0$.
Conversely: $x^2 + y^2 = 0 \Longrightarrow x = 0$ and $y = 0$. True, since $x^2 + y^2$ is non-negative and equals 0 if, and only if $x = y = 0$.

7.2.1. Exercises for class work / 3b.

$xy = xz \Longrightarrow y = z$. False, because if $x = 0, y = 2, z = 3$, then $0 \cdot 2 = 0 \cdot 3$, but $2 \neq 3$.
Conversely: $y = z \Longrightarrow xy = xz$. True.

7.2.1. Exercises for class work / 3c.

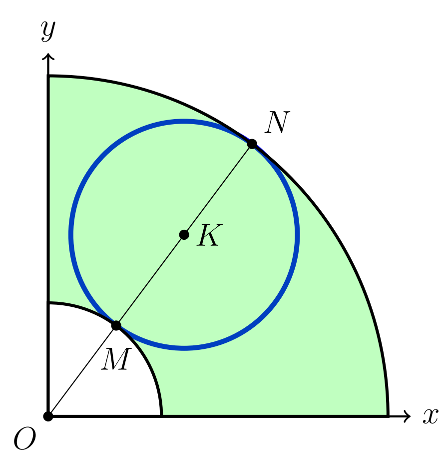
$x > y^2 \Longrightarrow x > 0$. True, because $x > y^2 \geq 0$.
Conversely: $x > 0 \Longrightarrow x > y^2$. False, because if $x = 1, y = 2$, then $1 > 0$, but $1 \not> 2^2$.

7.2.1. Exercises for class work / 3d.

$x^2 + y^2 = 12x + 16y - 75 \Longrightarrow 25 \leq x^2 + y^2 \leq 100$.
Transform the expression on the left side of the implication:

$$\begin{aligned} x^2 + y^2 &= 12x + 16y - 75 \\ x^2 - 12x + y^2 - 16y + 75 &= 0 \\ (x-6)^2 + (y-8)^2 - 25 &= 0 \\ (x-6)^2 + (y-8)^2 &= 5^2 \end{aligned}$$

This is a circle with center $(6, 8)$ and radius 5 (blue line of the circle in the figure), while the right side of the implication is a sector of the circle bounded by circles with radius 5 and 15 (marked in green on the figure, only in the first plane quarter). Based on these, the implication states that the points of the blue circle should fall into the green area. We can prove it by counting the number of intersecting points the circles with equations $x^2+y^2 = 25$ and $(x-6)^2+(y-8)^2 = 25$ have. Solving the system of equations obtained from the two equations, we get that the only solution is $x = 3$ and $y = 4$, so the two circles touch each other. Similarly, the circles with equations $x^2 + y^2 = 225$ and $(x - 6)^2 + (y - 8)^2 = 25$ also touch each other at the point $x = 9$ and $y = 12$. So the blue circle falls entirely on the green area, so the implication is true.



Conversely: $25 \le x^2+y^2 \le 100 \Longrightarrow x^2+y^2 = 12x+16y-75$, that is, every point of the green area is on the blue circle. Of course, this is false, for example if $x=10$ and $y=0$, then $25 \le 10^2+0^2 \le 225$, $10^2+0^2=100$ and $12\cdot 10+16\cdot 0-75=45$, but $100 \ne 45$.

7.2.1. Exercises for class work / 5a.

$\forall n \in \mathbb{N}^+ : \dfrac{1}{n} < 0.01$. This if false, for example if $n=2$, then $\dfrac{1}{2} \not< 0.01$

Negation: $\neg\left(\forall n \in \mathbb{N}^+ : \dfrac{1}{n} < 0.01\right) = \exists n \in \mathbb{N}^+ : \neg\left(\dfrac{1}{n} < 0.01\right) = \exists n \in \mathbb{N}^+ : \dfrac{1}{n} \ge 0.01$. This is true, there exists a natural number like this, for example $n=2$.

7.2.1. Exercises for class work / 5b.

$\exists n \in \mathbb{N}^+ : \dfrac{1}{n} < 0.01$. This is true because there exists such a natural number, for example: $n=101$.

Tagadása: $\neg\left(\exists n \in \mathbb{N}^+ : \dfrac{1}{n} < 0.01\right) = \forall n \in \mathbb{N}^+ : \neg\left(\dfrac{1}{n} < 0.01\right) = \forall n \in \mathbb{N}^+ : \dfrac{1}{n} \ge 0.01$. This is false, for example if $n=101$, then $\dfrac{1}{101} \not\ge 0.01$.

7.2.1. Exercises for class work / 5c.

This is an open statement because the parameter N is not fixed. (e.g. $N=20$ or $N=200$) Under different conditions of N, our statement can be both true and false.

7.2.1. Exercises for class work / 5d.

$\exists n \in \mathbb{N}^+ : \forall n \ge N : \dfrac{1}{n} < 0.01$. This statement is true, because, for example if $N=1000$, then for all values of n, which are grater or equal to 1000, we have that $\dfrac{1}{n} < 0.01$.

Negation:

$$\neg\left(\exists n \in \mathbb{N}^+ : \forall n \ge N : \frac{1}{n} < 0.01\right) = \forall n \in \mathbb{N}^+ : \neg\left(\forall n \ge N : \frac{1}{n} < 0.01\right) =$$
$$= \forall n \in \mathbb{N}^+ : \exists n \ge N : \neg\left(\frac{1}{n} < 0.01\right) = \forall n \in \mathbb{N}^+ : \exists n \ge N : \frac{1}{n} \ge 0.01.$$

Of course, this is false, because in the case, when $N=101$, if $n \ge 101$, then $\dfrac{1}{n} < 0.01$.

$\boxed{\text{7.2.1. Exercises for class work / 6a.}}$

$\exists n \in \mathbb{N} : \dfrac{n^2}{10n-7} > 100$. The statement is true, because according to the OPU-estimation:

$$\frac{n^2}{10n-7} > \frac{n^2}{10n} = \frac{n}{10} > 100.$$

From that, we know that $n > 1000$, i.e. $n = 1001$ is a good choice of n.
Negation: $\forall n \in \mathbb{N} : \dfrac{n^2}{10n-7} \leq 100$. This is false, e.g. if $n = 1001$.

$\boxed{\text{7.2.1. Exercises for class work / 6b.}}$

$\forall n \in \mathbb{N} : \dfrac{n^2}{10n-7} > 100$. The statement is false, for example if $n = 1$, then $\dfrac{n^2}{10n-7} = \dfrac{1}{3} \not> 100$.
Negation: $\exists n \in \mathbb{N} : \dfrac{n^2}{10n-7} \leq 100$, ehich is true, $n = 1$ is a good counter example for this.

$\boxed{\text{7.2.1. Exercises for class work / 6c.}}$

$\exists N \in \mathbb{N} : \forall n \geq N : \dfrac{n^2}{10n-7} > 100$, which is true, according to subtask a), because we know, that $\dfrac{n^2}{10n-7} > \dfrac{n}{10}$, that is, if we choose $N = 1001$ then for all $n \geq N$ the statement will be true.
Negation: $\forall N \in \mathbb{N} : \exists n \geq N : \dfrac{n^2}{10n-7} \leq 100$. Of course, this is false, $N = 1001$ is a good counter example for this.

$\boxed{\text{7.2.1. Exercises for class work / 7b.}}$

$\exists N \in \mathbb{N} : \forall n \geq N : \dfrac{2n^3+3}{n^5-3n^4-7n^3+2n^2-10n+1} < 0.05$. Using the OPU-estimation:

$$\frac{2n^3+3}{n^5-3n^4-7n^3+2n^2-10n+1} < \frac{2n^3+3n^3}{n^5-3n^4-7n^3-10n} = \frac{5n^3}{n^5-(3n^4+7n^3+10n)} <$$
$$< \frac{5n^3}{n^5-(3n^4+7n^4+10n^4)} = \frac{5n^3}{n^5-20n^4} = \frac{5n^3}{\frac{1}{2}n^5 + n^4\left(\frac{1}{2}n-20\right)} < \frac{5n^3}{\frac{1}{2}n^5} = \frac{10}{n^2}, \text{if } n > 40.$$

According to this, the statement is true, because

$$\frac{2n^3+3}{n^5-3n^4-7n^3+2n^2-10n+1} < \frac{10}{n^2} < 0.05 \Longrightarrow 200 < n^2 \Longrightarrow 15 \leq n.$$

It is important to note that although we got 15 for n in the last estimate, we have to be very careful that we can only guarantee the OPU-estimate from $n = 40$, so $N = \ max(15, 40) = 40$ is a good choice.

Negation of the original statement: $\forall N \in \mathbb{N} : \exists n \geq N : \dfrac{2n^3+3}{n^5-3n^4-7n^3+2n^2-10n+1} \geq 0.05$.

7.2.1. Exercises for class work / 7c.

$\exists N \in \mathbb{N} : \forall n \geq N : \dfrac{2n^6-20n^5+3n^4-6n^3+3}{13n^3+100n^2+200} > 230$. Using the OPL-estimation:

$$\frac{2n^6-20n^5+3n^4-6n^3+3}{13n^3+100n^2+200} > \frac{2n^6-20n^5-6n^3}{13n^3+100n^3+200n^3} = \frac{2n^6-(20n^5+6n^3)}{313n^3} >$$
$$> \frac{2n^6-26n^5}{313n^3} = \frac{n^6+n^6-26n^5}{313n^3} = \frac{n^6+n^5(n-26)}{313n^3} > \frac{n^6}{313n^3} = \frac{n^3}{313}, \text{if } n > 26.$$

According to this, the statement is true, because

$$\frac{2n^6-20n^5+3n^4-6n^3+3}{13n^3+100n^2+200} > \frac{n^3}{313} > 230 \Longrightarrow n^3 > 71990 \Longrightarrow n \geq 42.$$

$N = \max(26, 42) = 42$ is a good choice.

Negation of the original statement: $\forall N \in \mathbb{N} : \exists n \geq N : \dfrac{2n^6-20n^5+3n^4-6n^3+3}{13n^3+100n^2+200} \leq 230$.
